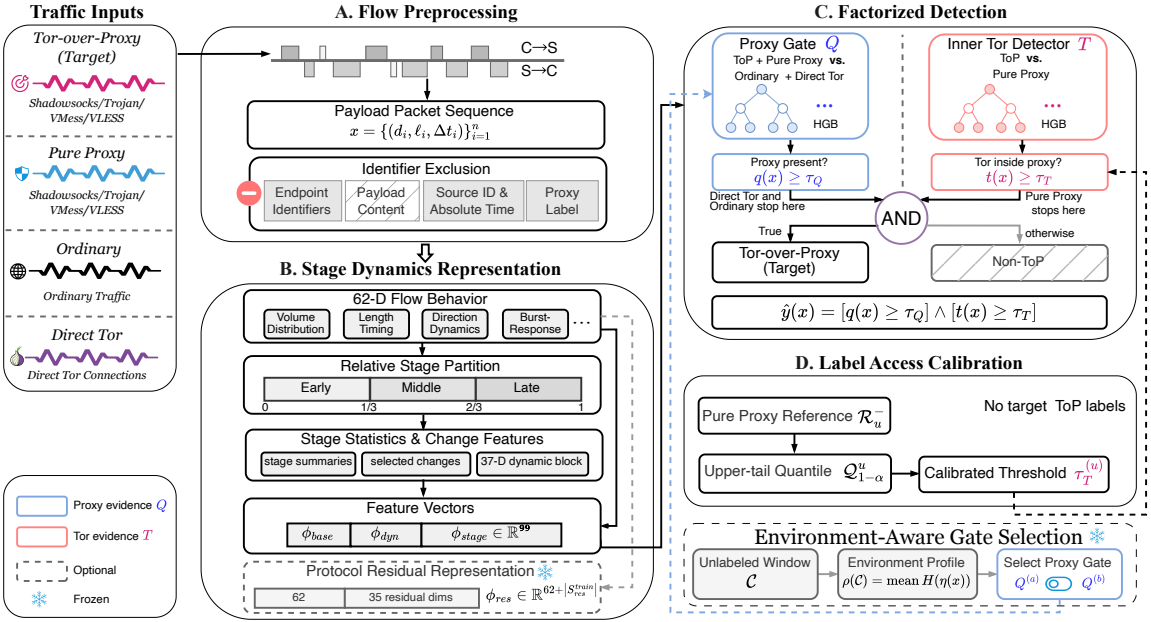

Traffic Inputs
Tor-over-Proxy (Target)
Shadowsocks/Trojan/ VMess/VLESS
Pure Proxy
Shadowsocks/Trojan/ VMess/VLESS
Ordinary
Ordinary Traffic
Direct Tor
Direct Tor Connections
Proxy evidence Q
Tor evidence T
Optional
Frozen
A. Flow Preprocessing
C→S
S→C
Payload Packet Sequence
$x = \{(d_i, \ell_i, \Delta t_i)\}_{i=1}^{n}$
Identifier Exclusion
Endpoint Identifiers
Payload Content
Source ID & Absolute Time
Proxy Label
B. Stage Dynamics Representation
62-D Flow Behavior
Volume Distribution
Length Timing
Direction Dynamics
Burst-Response
Relative Stage Partition
Early
Middle
Late
0
1/3
2/3
1
Stage Statistics & Change Features
stage summaries
selected changes
37-D dynamic block
Feature Vectors
ϕ_{base}
ϕ_{dyn}
$\phi_{stage} \in \mathbb{R}^{99}$
Protocol Residual Representation
62
35 residual dims
$\phi_{res} \in \mathbb{R}^{62+|S_{res}^{train}|}$
C. Factorized Detection
Proxy Gate Q
ToP + Pure Proxy vs. Ordinary + Direct Tor
HGB
Inner Tor Detector T
ToP vs. Pure Proxy
HGB
Proxy present?
$q(x) \geq \tau_Q$
Tor inside proxy?
$t(x) \geq \tau_T$
Direct Tor and Ordinary stop here
Pure Proxy stops here
AND
True
otherwise
Tor-over-Proxy (Target)
Non-ToP
$\hat{y}(x) = [q(x) \geq \tau_Q] \wedge [t(x) \geq \tau_T]$
D. Label Access Calibration
No target ToP labels
Pure Proxy Reference $\mathcal{R}_u^-$
Upper-tail Quantile $\mathcal{Q}_{1-\alpha}^u$
Calibrated Threshold $\tau_T^{(u)}$
Environment-Aware Gate Selection
Unlabeled Window $\mathcal{C}$
Environment Profile $\rho(\mathcal{C}) = \text{mean}\, H(\eta(x))$
Select Proxy Gate
$Q^{(a)}$
$Q^{(b)}$